

FaceSense: Real-Time Face Recognition and Facial Emotion Analysis

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Abstract—Face recognition and facial emotion analysis are increasingly central to modern artificial intelligence applications, ranging from security systems to human–computer interaction. However, building a system capable of performing both tasks in real time with high accuracy remains a significant research challenge, particularly given the difficulty of classifying visually ambiguous emotions from low-resolution images. In this paper, we propose FaceSense, a real-time intelligent system that combines face detection, identity recognition, and facial emotion classification within a unified pipeline. The system is built using Python, OpenCV, DeepFace, and Streamlit for live webcam interaction. For emotion recognition, three deep learning models are evaluated: a convolutional neural network trained from scratch, and two transfer learning approaches based on MobileNetV2 and ResNet50, all trained on the FER2013 dataset. Experimental results show that the fine-tuned ResNet50 model achieves the best classification accuracy of 63.84%, outperforming both the CNN from scratch (60.39%) and the MobileNetV2 fine-tuned model (61.28%), demonstrating the effectiveness of transfer learning in improving recognition performance and reducing training complexity.

Keywords: Deep Learning, Face Recognition, Emotion Recognition, Transfer Learning, CNN, Computer Vision, DeepFace, OpenCV

I. INTRODUCTION

Artificial intelligence and deep learning have profoundly transformed the landscape of computer vision over the past decade. Among the most active research areas are face recognition and facial emotion analysis, which underpin a wide spectrum of applications including access control systems, healthcare monitoring, smart surveillance, and affective human–computer interaction [1]. The growing availability of powerful pre-trained models and large-scale facial datasets has further accelerated progress in this field, making it feasible to deploy sophisticated recognition systems even on resource-constrained devices.

Despite significant advances, building a unified system capable of performing both face recognition and real-time emotion classification with high accuracy remains a challenging problem. On one hand, face recognition must be robust to variations in lighting, pose, and partial occlusion. On the other hand, facial emotion recognition faces additional difficulties inherent to the FER2013 dataset, which contains low-resolution 48×48 grayscale images, severe class imbalance (the class *disgust* accounts for fewer than 500 samples versus over 7,000 for *happy*), and significant visual similarity between certain emotion categories such as *fear*, *sad*, and *neutral*.

Convolutional neural networks trained from scratch on such data tend to overfit or perform poorly on minority classes, while more powerful architectures trained on generic datasets (e.g., ImageNet) may not transfer perfectly to the subtle facial domain. There is therefore a need to systematically compare multiple deep learning strategies—from scratch training to fine-tuned transfer learning—on this specific task.

To address these challenges, this paper makes the following contributions: design and implementation of FaceSense, a real-time pipeline integrating face detection, identity recognition via FaceNet-based embeddings (DeepFace), and facial emotion classification from live webcam streams; a comprehensive evaluation of three deep learning architectures for emotion recognition on FER2013 including a CNN trained from scratch, MobileNetV2 with transfer learning and fine-tuning, and ResNet50 with transfer learning and fine-tuning; a systematic analysis of preprocessing strategies including normalization, augmentation, and class weighting to mitigate class imbalance and improve model generalization; and deployment of the full system through an interactive Streamlit web interface enabling real-time inference.

The remainder of this paper is organized as follows. Section II reviews related work on face recognition and facial emotion analysis. Section III describes the complete system methodology. Section IV presents results and discussion. Section V concludes the paper with directions for future work.

II. LITERATURE REVIEW

Facial emotion recognition (FER) has evolved considerably over the past two decades, transitioning from handcrafted feature approaches to deep learning solutions. Early methods relied on engineered descriptors such as Local Binary Patterns (LBP) [2] and Histogram of Oriented Gradients (HOG) combined with Support Vector Machines (SVM). While computationally efficient, these approaches struggle to capture the complex, non-linear facial deformations associated with subtle emotions.

The advent of Convolutional Neural Networks (CNNs) marked a turning point in FER. Goodfellow et al. [4] introduced the FER2013 dataset alongside a CNN baseline, establishing a benchmark that remains widely used. Subsequent work demonstrated that deeper architectures and data augmentation strategies could significantly improve performance.

However, training deep CNNs from scratch on FER2013 is limited by the relatively small size and high class imbalance of the dataset.

Transfer learning has emerged as a powerful technique to overcome dataset limitations. Architectures pre-trained on ImageNet, such as VGG16, MobileNetV2, and ResNet50, have been successfully fine-tuned for FER tasks. MobileNetV2 is particularly attractive for real-time applications due to its lightweight depthwise separable convolutions, offering a strong trade-off between accuracy and computational cost. ResNet50, introduced by He et al. and based on residual learning [5], provides deeper feature extraction through skip connections that prevent the vanishing gradient problem, enabling richer hierarchical representations of facial features.

For face recognition, FaceNet [2] introduced a unified embedding approach that maps face images to a compact Euclidean space where distances directly correspond to facial similarity. This embedding strategy, implemented in the DeepFace library, has become a standard baseline for open-set recognition tasks. Parkhi et al. [3] further demonstrated the effectiveness of very deep CNNs for face recognition on large-scale datasets.

In contrast to prior work that addresses recognition or emotion classification in isolation, our system integrates both tasks in a unified real-time pipeline, and systematically compares three training strategies—from-scratch CNN, MobileNetV2 transfer learning, and ResNet50 transfer learning with fine-tuning—on the same dataset and evaluation protocol. Recent research in facial emotion recognition has also explored large-scale datasets such as AffectNet [9] and RAF-DB [10], which provide more diverse facial expressions captured in unconstrained environments.

In addition, lightweight architectures such as MobileNetV2 [6] and deeper residual networks such as ResNet50 [7] have become strong baselines for FER due to their efficiency and strong visual feature extraction capabilities.

III. METHODOLOGY

A. System Overview

FaceSense is a real-time intelligent system that integrates face detection, identity recognition, and facial emotion classification within a unified pipeline. As illustrated in Fig. 1, the system consists of two parallel processing modules triggered by a common face detection stage. Module 1 handles identity recognition through facial embedding extraction and cosine similarity comparison. Module 2 performs emotion classification using deep convolutional neural networks trained on FER2013. The outputs of both modules are fused and displayed in real time through a Streamlit web interface. The system is implemented using Python, OpenCV, DeepFace, TensorFlow/Keras, and Streamlit.

B. Notation

Table I defines the mathematical symbols used throughout this section.

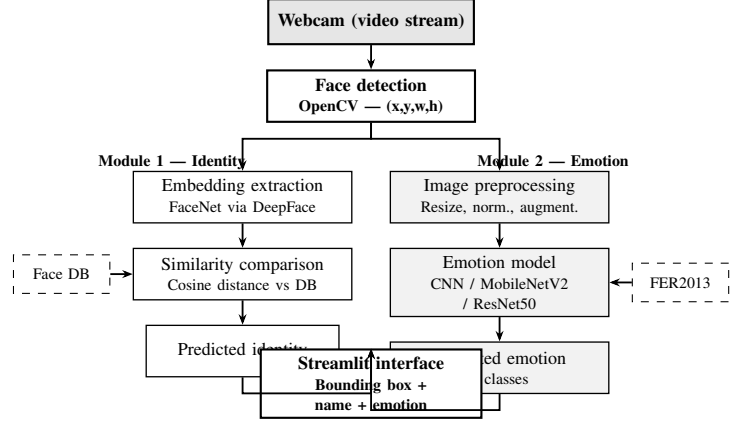


Fig. 1: Global architecture of the FaceSense system.

TABLE I: Mathematical Notation

Symbol	Definition
I	Input face image (matrix $H \times W \times C$)
E	Facial embedding vector $\in \mathbb{R}^n$
D	Database of known embeddings $\{E_1, \dots, E_k\}$
$d(A, B)$	Cosine distance between embeddings A and B
τ	Recognition decision threshold
$\hat{y}$	Predicted emotion class $\in \{0, \dots, 6\}$
$\hat{p}$	Softmax probability vector $\in \mathbb{R}^7$
x_{norm}	Pixel-normalized image in $[0, 1]$

C. Theoretical Background

1) *Face Embedding with FaceNet*: FaceSense relies on the FaceNet model [2], which maps a face image I to a compact embedding vector $E \in \mathbb{R}^n$ such that embeddings of the same individual cluster together while embeddings of different individuals are far apart. This approach enables open-set recognition without requiring model retraining when new individuals are added to the database.

2) *Cosine Distance*: Identity recognition uses cosine distance as the similarity metric:

$$d(A, B) = 1 - \frac{A \cdot B}{\|A\| \times \|B\|} \quad (1)$$

A face is recognized when $d(E, E_i) < \tau$; otherwise it is labeled as unknown. The predicted identity is selected as:

$$\hat{y}_{\text{id}} = \arg \min_i d(E, E_i), \quad \text{s.t. } d(E, E_i) < \tau \quad (2)$$

3) *Residual Learning*: ResNet50 is based on residual learning [4]. Instead of learning a direct mapping $H(x)$, each residual block learns a residual function:

$$F(x) = H(x) - x \implies H(x) = F(x) + x \quad (3)$$

The skip connection allows gradients to propagate through very deep networks, preventing the vanishing gradient problem and enabling richer hierarchical feature extraction compared to shallow architectures.

D. Identity Recognition Module (Module 1)

1) *Face Detection*: The webcam stream is processed frame by frame using OpenCV and DeepFace. Each detected facial region is represented as a bounding box (x, y, w, h) , where (x, y) is the top-left corner, w is the width, and h is the height.

2) *Known Faces Dataset*: The identity recognition module uses a custom dataset of known individuals. Each person's images are stored in a dedicated folder:

```
dataset/  
  person1/  img1.jpg  
  person2/  img1.jpg
```

At application startup, a FaceNet embedding is pre-computed and cached for each stored image to avoid redundant computation during inference.

3) *Embedding Extraction*: Instead of directly comparing raw images, DeepFace converts each detected face into a numerical embedding vector representing high-level facial features:

$$E = [e_1, e_2, \dots, e_n] \quad (4)$$

4) *Recognition Decision*: The system compares the live embedding E against all stored embeddings in D using Eq. (1). If the similarity score is below the threshold τ , the face is recognized and the predicted identity is displayed using OpenCV. Otherwise, the face is labeled as “Unknown.”

E. FER2013 Dataset and Exploratory Analysis

For the facial emotion recognition task, we used the FER2013 dataset [4], a widely used benchmark containing grayscale images of size 48×48 pixels categorized into seven emotion classes (Table II).

TABLE II: FER2013 Emotion Classes

Class	Emotion
0	Angry
1	Disgust
2	Fear
3	Happy
4	Neutral
5	Sad
6	Surprise

A significant class imbalance is observed: *disgust* contains fewer than 500 samples while *happy* contains over 7,000. This imbalance is addressed during training through class weighting. An exploratory data analysis was performed before training to understand the structure and quality of the dataset, as shown in Figs. 2 and 3.

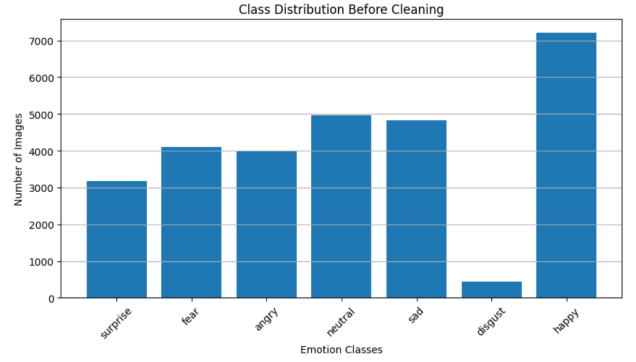


Fig. 2: Class distribution before and after preprocessing.

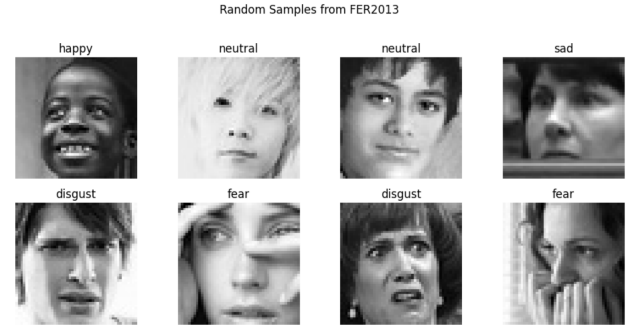


Fig. 3: Sample images from the FER2013 dataset.

F. Preprocessing Pipeline

A unified preprocessing pipeline was applied to all images before training.

Resizing. Images were resized to the input dimensions required by each architecture: 48×48 for the CNN from scratch, 160×160 for MobileNetV2, and 224×224 for ResNet50. Grayscale images were converted to RGB for compatibility with ImageNet-pretrained models.

Normalization. Pixel values originally in $[0, 255]$ were normalized to $[0, 1]$ using:

$$x_{\text{norm}} = \frac{x}{255} \quad (5)$$

Normalization reduces numerical instability and accelerates gradient-based optimization. For transfer learning models, architecture-specific preprocessing functions were additionally applied to match ImageNet statistics.

Data Augmentation. To improve generalization and reduce overfitting, the following transformations were applied during training: horizontal flipping, random rotation, random zoom, random translation, and random contrast adjustment (the last two for transfer learning models only). These preprocessing strategies are consistent with previous deep learning studies in computer vision and emotion recognition, where normalization and augmentation play a key role in improving model robustness and reducing overfitting [8].

G. Deep Learning Models for Emotion Classification

Three architectures were evaluated under identical experimental conditions.

1) *CNN Trained from Scratch*: A custom CNN receives $48 \times 48 \times 1$ grayscale images and outputs a probability vector $\hat{p} \in \mathbb{R}^7$. The architecture consists of three convolutional blocks (Conv2D, Batch Normalization, MaxPooling, Dropout) followed by a fully connected head (Dense 256, Dropout, Dense 7 Softmax), as detailed in Table III.

TABLE III: CNN From Scratch Architecture

Layer	Description
Input	$48 \times 48 \times 1$ grayscale
Conv2D	32 filters, 3×3 , ReLU
BatchNorm + Pool + Dropout	Block 1
Conv2D	64 filters, 3×3 , ReLU
BatchNorm + Pool + Dropout	Block 2
Conv2D	128 filters, 3×3 , ReLU
BatchNorm + Pool + Dropout	Block 3
Dense	256 neurons, ReLU
Dropout	Regularization
Dense	7 neurons, Softmax

2) *MobileNetV2 with Transfer Learning and Fine-Tuning*: MobileNetV2, pre-trained on ImageNet, is used as a lightweight feature extractor. The original classification head is replaced by: Global Average Pooling, Batch Normalization, Dropout, Dense (128, ReLU), Dropout, Dense (7, Softmax) (Table IV). Training proceeds in two stages: (1) the convolutional base is frozen and only the custom head is trained; (2) the final layers are unfrozen and fine-tuned with $lr = 10^{-5}$. MobileNetV2 was introduced by Sandler et al. [6] and is widely adopted for efficient computer vision applications due to its low computational cost and strong performance on embedded and real-time systems.

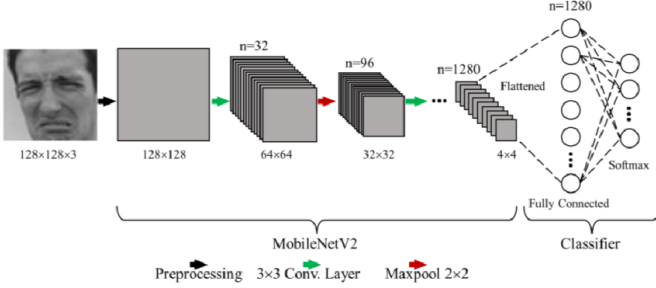


Fig. 4: Workflow of the MobileNetV2-based emotion recognition model.

TABLE IV: MobileNetV2 Transfer Learning Architecture

Layer	Description
Input	$160 \times 160 \times 3$ RGB
Data Augmentation	Random transformations
Preprocessing	MobileNetV2 preprocessing
MobileNetV2 base	Pre-trained on ImageNet
GlobalAvgPool2D	Feature reduction
BatchNorm + Dropout	Regularization
Dense	128 neurons, ReLU
Dropout	Regularization
Dense	7 neurons, Softmax

3) *ResNet50 with Transfer Learning and Fine-Tuning*: ResNet50, also pre-trained on ImageNet, is used as a deeper feature extractor. The same two-stage training strategy is applied with initial $lr = 10^{-4}$ and fine-tuning $lr = 10^{-5}$. The larger input resolution (224×224) enables deeper layers to capture more detailed facial patterns (Table V). ResNet50 was proposed by He et al. [7] and introduced residual learning, which enables training deeper neural networks while mitigating the vanishing gradient problem.

TABLE V: ResNet50 Transfer Learning Architecture

Layer	Description
Input	$224 \times 224 \times 3$ RGB
Data Augmentation	Random transformations
Preprocessing	ResNet50 preprocessing
ResNet50 base	Pre-trained on ImageNet
GlobalAvgPool2D	Feature reduction
BatchNorm + Dropout	Regularization
Dense	256 neurons, ReLU
Dropout	Regularization
Dense	7 neurons, Softmax

H. Training Configuration and Evaluation Metrics

All models share the training configuration shown in Table VI.

TABLE VI: Training Configuration

Parameter	Value
Optimizer	Adam
Loss function	Sparse Categorical Crossentropy
Epochs (CNN)	100
Batch size	32
Early stopping	Enabled (restore best weights)
ReduceLROnPlateau	Enabled
Class weighting	Enabled

Models are evaluated using accuracy, precision, recall, and F1-score:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

IV. RESULTS AND DISCUSSION

A. CNN From Scratch

The CNN trained from scratch achieved 60.39% accuracy on the FER2013 test set (Table VII). The close values between accuracy, precision, and recall confirm that class weighting successfully mitigated the impact of class imbalance.

TABLE VII: CNN From Scratch — Global Results

Metric	Value
Accuracy	60.39%
Precision	60.41%
Recall	60.39%
Loss	1.058

Per-class analysis (Table VIII) shows that *happy* (F1=0.84) and *surprise* (F1=0.75) are easiest to classify due to their visually distinctive features. Conversely, *disgust* (F1=0.13) suffers from critically low recall, reflecting its severe under-representation in the dataset. Similarly, *fear* can be confused with *sad*, *neutral*, or *surprise* because these emotions share similar facial patterns.

TABLE VIII: CNN From Scratch — Per-Class Results

Class	Precision	Recall	F1
Angry	0.50	0.53	0.52
Disgust	0.73	0.07	0.13
Fear	0.47	0.24	0.32
Happy	0.83	0.85	0.84
Neutral	0.48	0.71	0.57
Sad	0.48	0.45	0.46
Surprise	0.75	0.76	0.75

The validation accuracy (62.54%) is slightly higher than the training accuracy (58.32%), explained by the use of dropout and data augmentation during training. The absence of a large gap between the two values confirms that the model does not suffer from severe overfitting.

B. MobileNetV2

Initial transfer learning achieved 60.35% accuracy, comparable to the CNN from scratch but with faster convergence. Fine-tuning improved accuracy to 61.28% with a reduced loss of 1.045 (Table IX). MobileNetV2 offers a strong compromise between computational efficiency, inference speed, model size, and classification performance, making it suitable for real-time deployment. The performance plateau around 61% suggests that the primary bottleneck lies in the dataset rather than the architecture.

TABLE IX: MobileNetV2 Results

Stage	Accuracy	Loss
Transfer learning	60.35%	1.060
Fine-tuning	61.28%	1.045

C. ResNet50

ResNet50 with fine-tuning achieved the best performance among all evaluated models: 63.84% accuracy and 0.982 loss (Table X). The deeper residual architecture enabled richer extraction of hierarchical facial features, translating into better emotion discrimination. However, ResNet50 also introduces higher computational cost, larger model size, and slower inference speed compared to MobileNetV2.

TABLE X: ResNet50 Results

Metric	Value
Test Accuracy	63.84%
Test Loss	0.982

D. Comparative Analysis

Table XI summarizes the performance of all evaluated models. ResNet50 consistently achieves the highest accuracy and lowest loss, confirming that transfer learning with fine-tuning from a deep architecture is the most effective strategy for the FER2013 task.

TABLE XI: Comparison of All Models

Model	Accuracy	Loss
CNN from scratch	60.39%	1.058
MobileNetV2 (TL)	60.35%	1.060
MobileNetV2 (fine-tuned)	61.28%	1.045
ResNet50 (TL + fine-tuned)	63.84%	0.982

The results also reveal that the primary performance bottleneck is the FER2013 dataset itself—its low resolution (48×48), class imbalance, and label ambiguity—rather than the model architectures. Class weighting, data augmentation, and careful preprocessing partially alleviate these constraints but cannot fully compensate for inherent dataset limitations.

V. DEPLOYMENT

The complete FaceSense pipeline is deployed as a Streamlit web application enabling real-time interaction through a standard webcam. Upon launching the application, the system executes the following operations for each captured frame:

- 1) Capture the webcam frame
- 2) Detect facial regions using OpenCV and DeepFace
- 3) Extract FaceNet embeddings and predict identity (Module 1)
- 4) Preprocess the face and classify the emotion (Module 2)
- 5) Overlay the predicted identity and emotion label on the bounding box
- 6) Display the annotated frame in the Streamlit interface

The deployment interface was developed using Streamlit [14], while image processing operations relied on OpenCV [11]. Model training and inference were implemented using TensorFlow [12] and Keras [13].

VI. CONCLUSION

This paper presented FaceSense, a real-time intelligent system combining face detection, identity recognition, and facial emotion classification within a unified deep learning pipeline. Three model architectures were systematically compared for emotion recognition on the FER2013 dataset under identical experimental conditions. Experimental results demonstrate that transfer learning with fine-tuning consistently outperforms training from scratch: ResNet50 achieved the best accuracy (63.84%), followed by MobileNetV2 fine-tuned (61.28%) and the CNN from scratch (60.39%). The primary performance bottleneck is identified as the FER2013 dataset itself, due to its low resolution, class imbalance, and label ambiguity.

Future work could explore: (1) training on larger and more balanced datasets such as AffectNet or RAF-DB; (2) multi-task learning jointly optimizing recognition and emotion classification; (3) GPU-accelerated inference for edge device deployment; and (4) extension to multi-face real-time scenarios.

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